



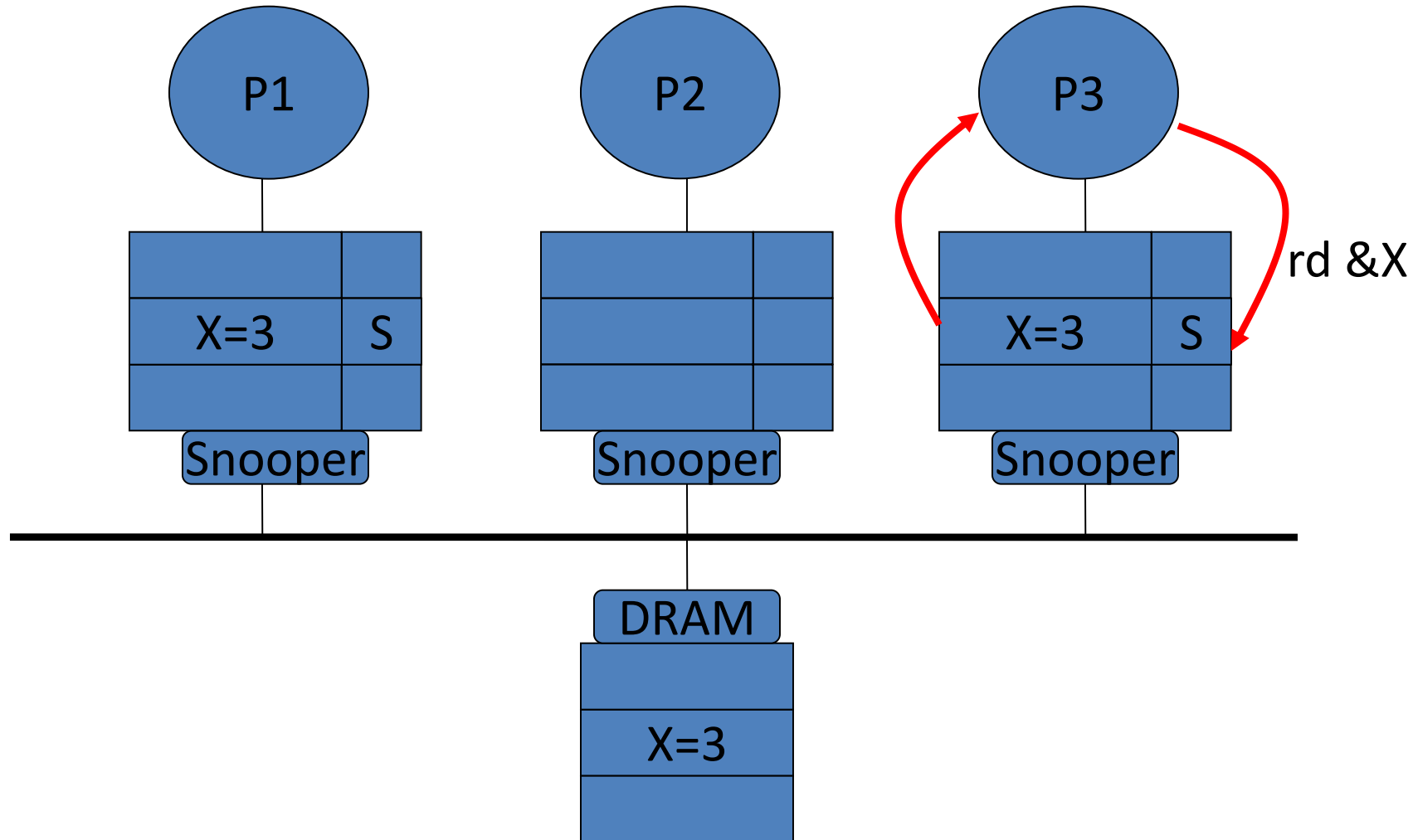
CS683: Advanced Computer Architecture

Lecture 13: Cache Coherence-II

<https://www.cse.iitb.ac.in/~biswa/courses/>

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New structure in the hierarchy: Snooper



Summary

Proc Action	State P1	State P2	State P3	Bus Action	Data From
R1	-	-	-	BusRd	Mem
W1	-	-	-	BusRdX	Mem
R3	-	-	-	BusRd	P1 cache
W3	-	-	-	BusRdX	Mem
R1	-	-	-	BusRd	P3 cache
R3	-	-	-	-	-
R2	-	-	-	BusRd	Mem

Summary

Proc Action	State P1	State P2	State P3	Bus Action	Data From
R1	S	-	-	BusRd	Mem
W1	M	-	-	BusRdX	Mem
R3	S	-	S	BusRd	P1 cache
W3	I	-	M	BusRdX	Mem
R1	S	-	S	BusRd	P3 cache
R3	S	-	S	-	-
R2	S	S	S	BusRd	Mem

More on coherence protocols

- MESI
- MOESI (Owner cache line that will initiate cache to cache transfer)

Problem with MSI protocol

Rd, Wr sequence incurs 2 transactions

- even when no one sharing (e.g., serial program!)
- BusRd (I- $\rightarrow$ S) followed by BusRdX (S- $\rightarrow$ M)

- *In general, penalizing serial programs is unacceptable*

Solution – MESI

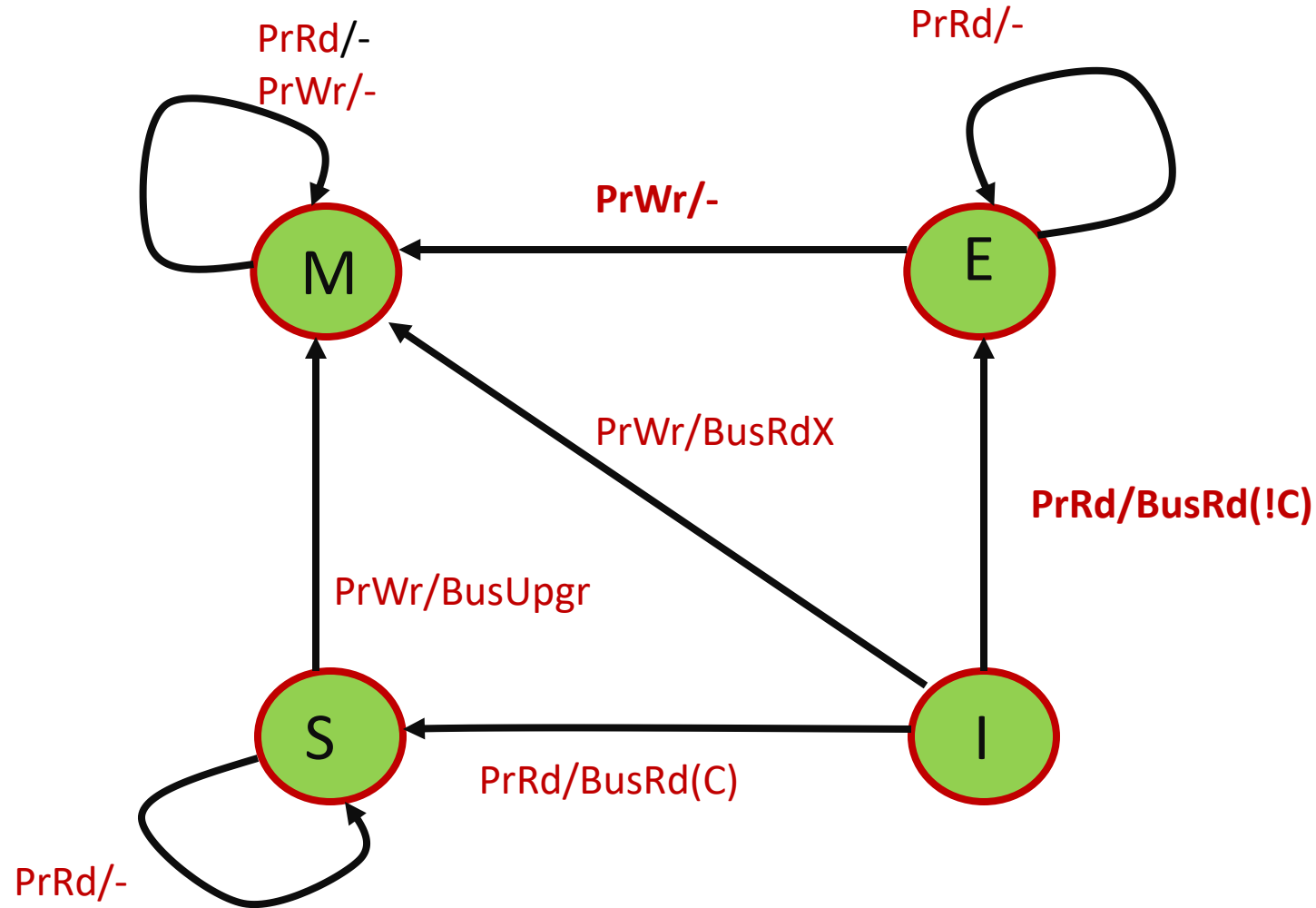
Add **E***xclusive* state:

- Invalid
- modified (dirty)
- shared (two or more caches may have copies)
- Exclusive: (only this cache has *clean* copy, same value as in memory)

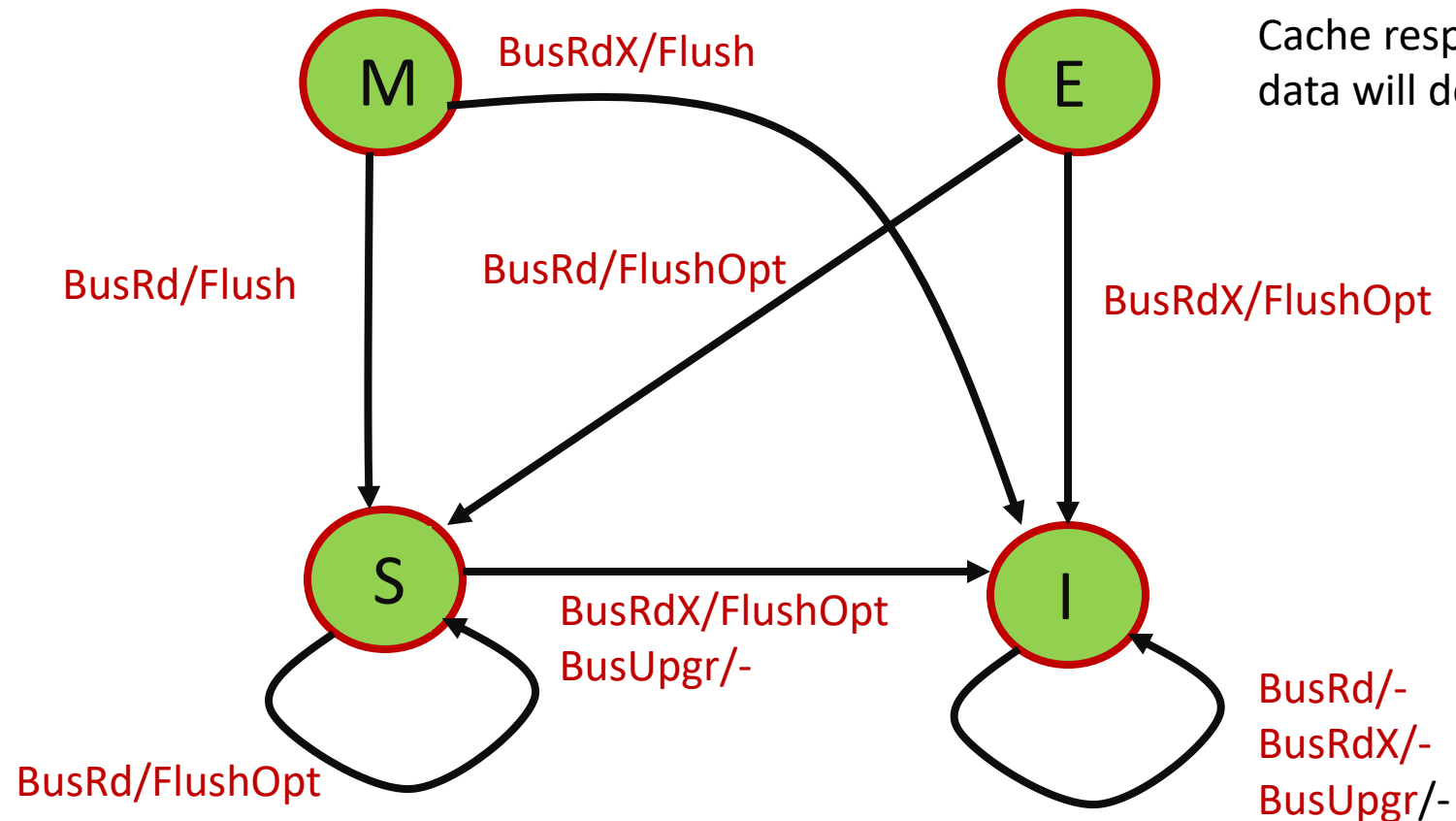
How to decide $I \rightarrow E$ or $I \rightarrow S$?

- Need to check whether someone else has copy
- a separate signal on bus: *wired-or* line asserted in response to BusRd. Call it C = “copy exists”

State Transition Diagram(Core Side)

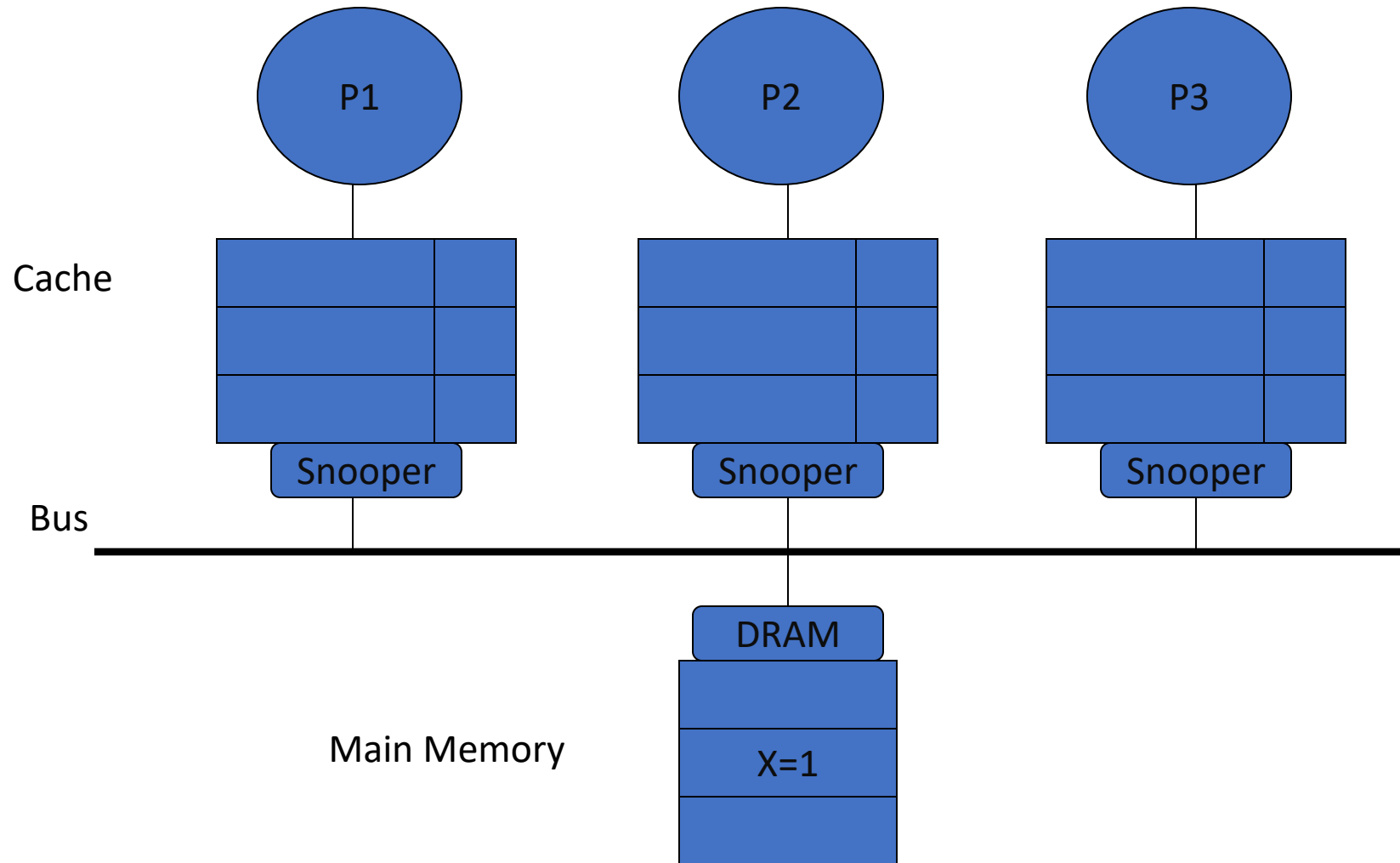


State Transition Diagram(Bus Side)

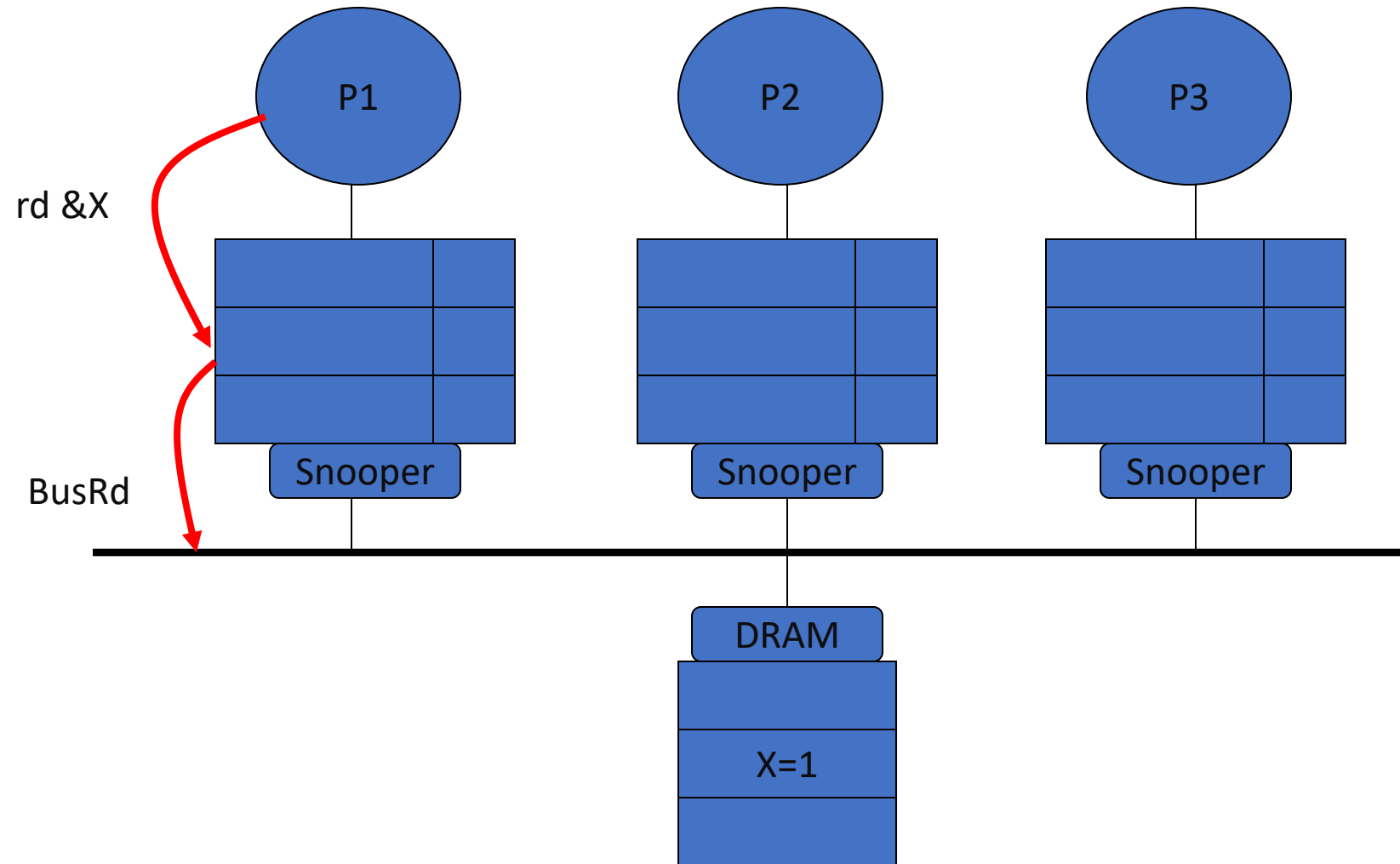


Flushopt is flush only if cache to cache sharing is used. Only the Cache responsible for supplying the data will do the flush.

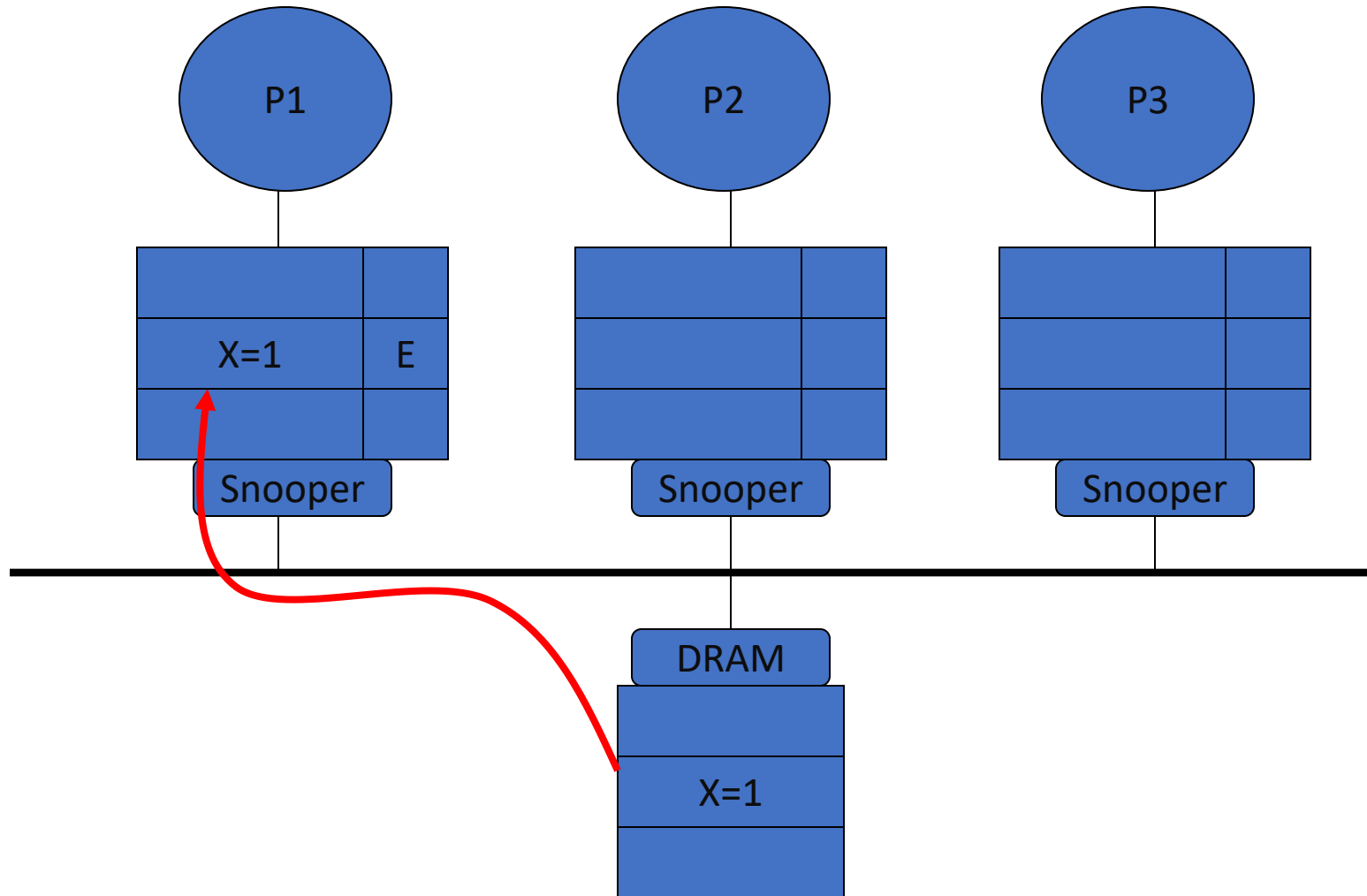
MESI



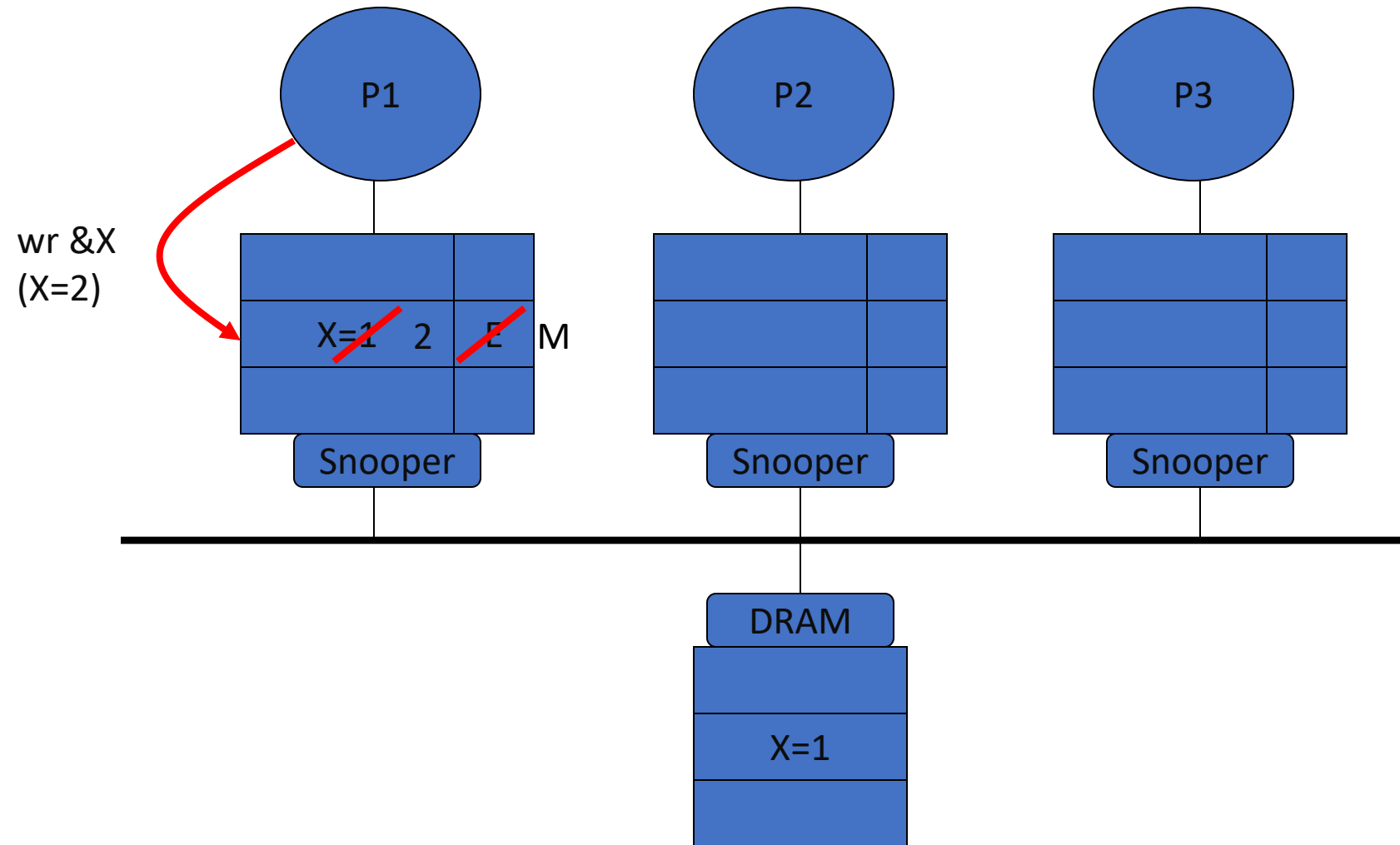
MESI



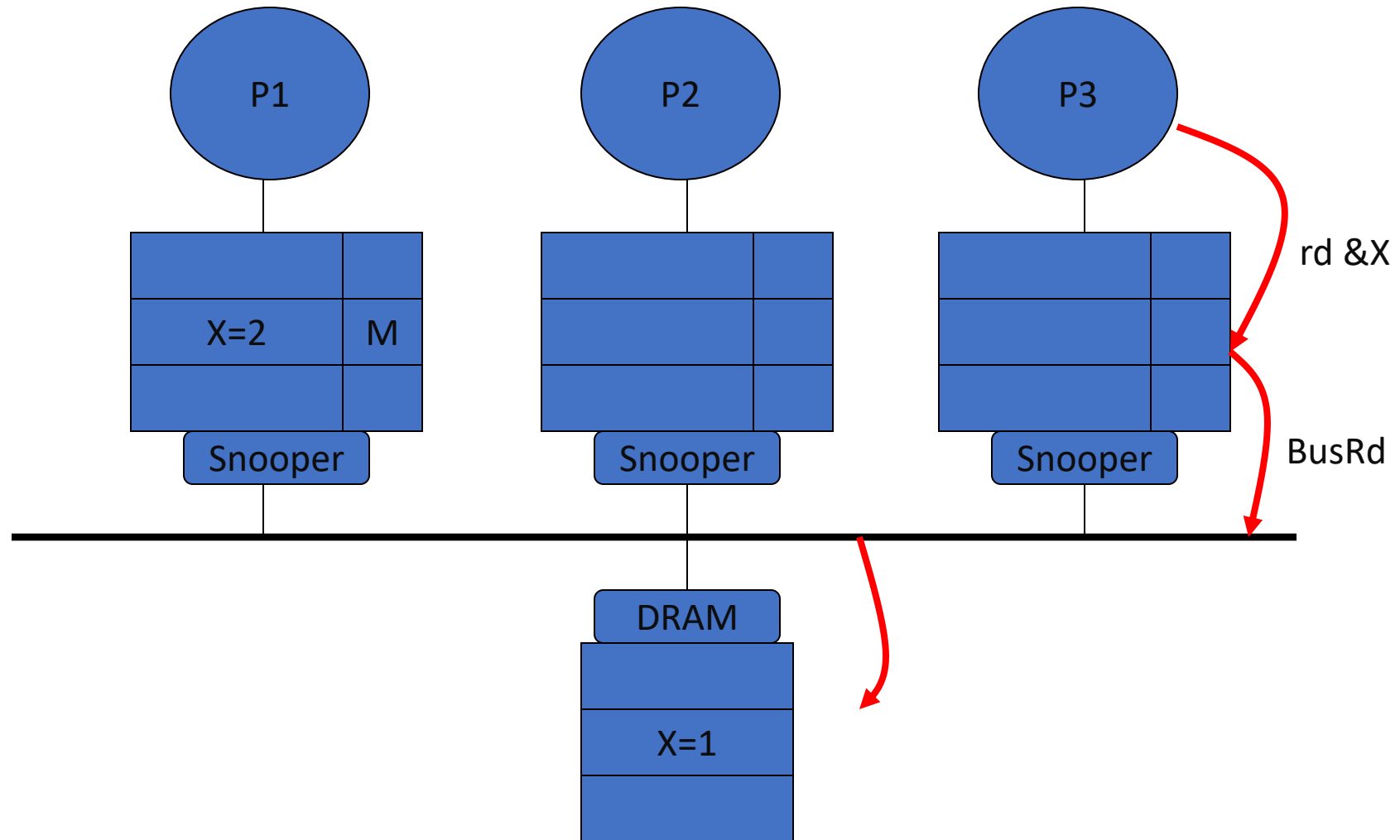
MESI



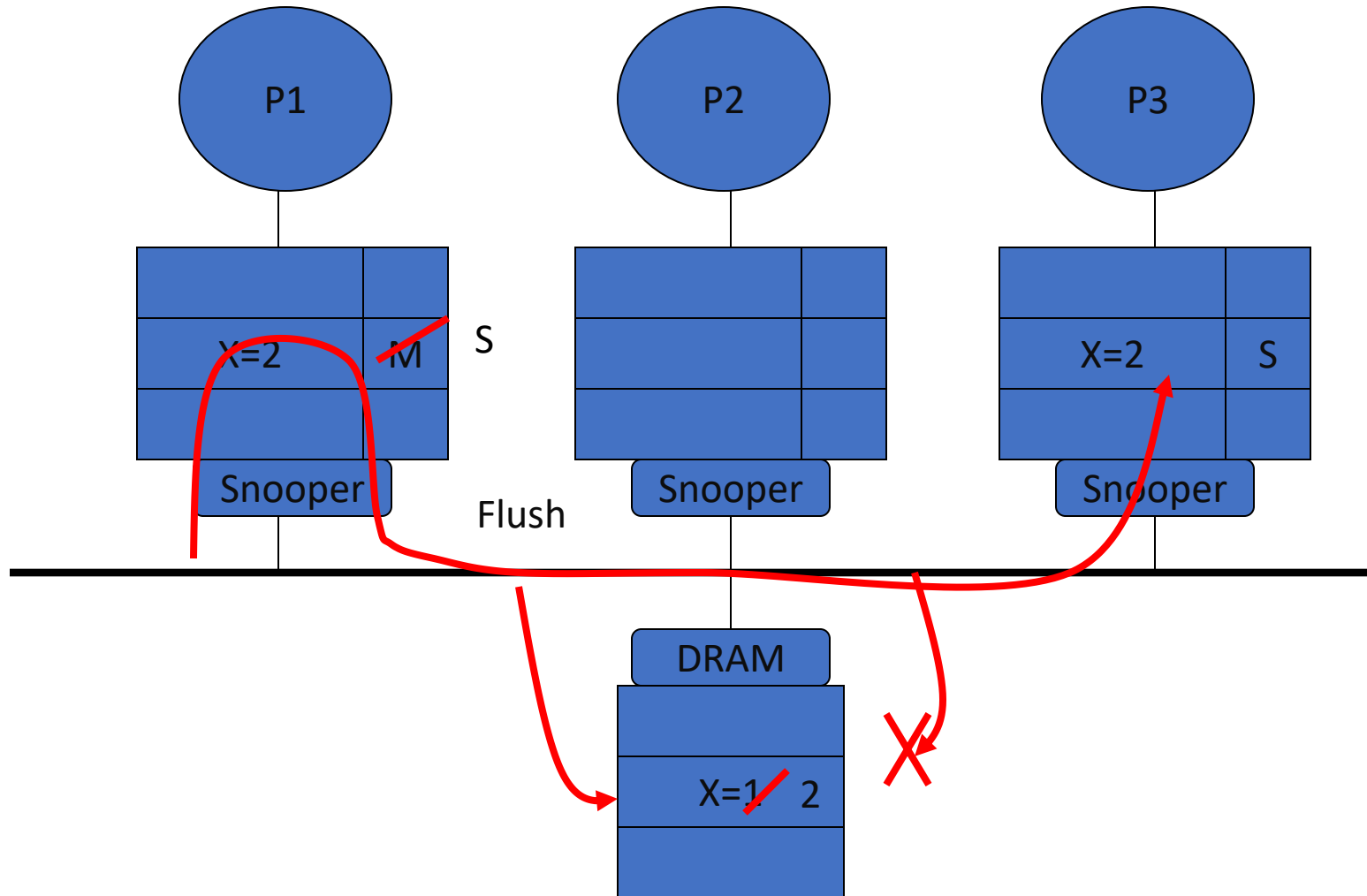
MESI



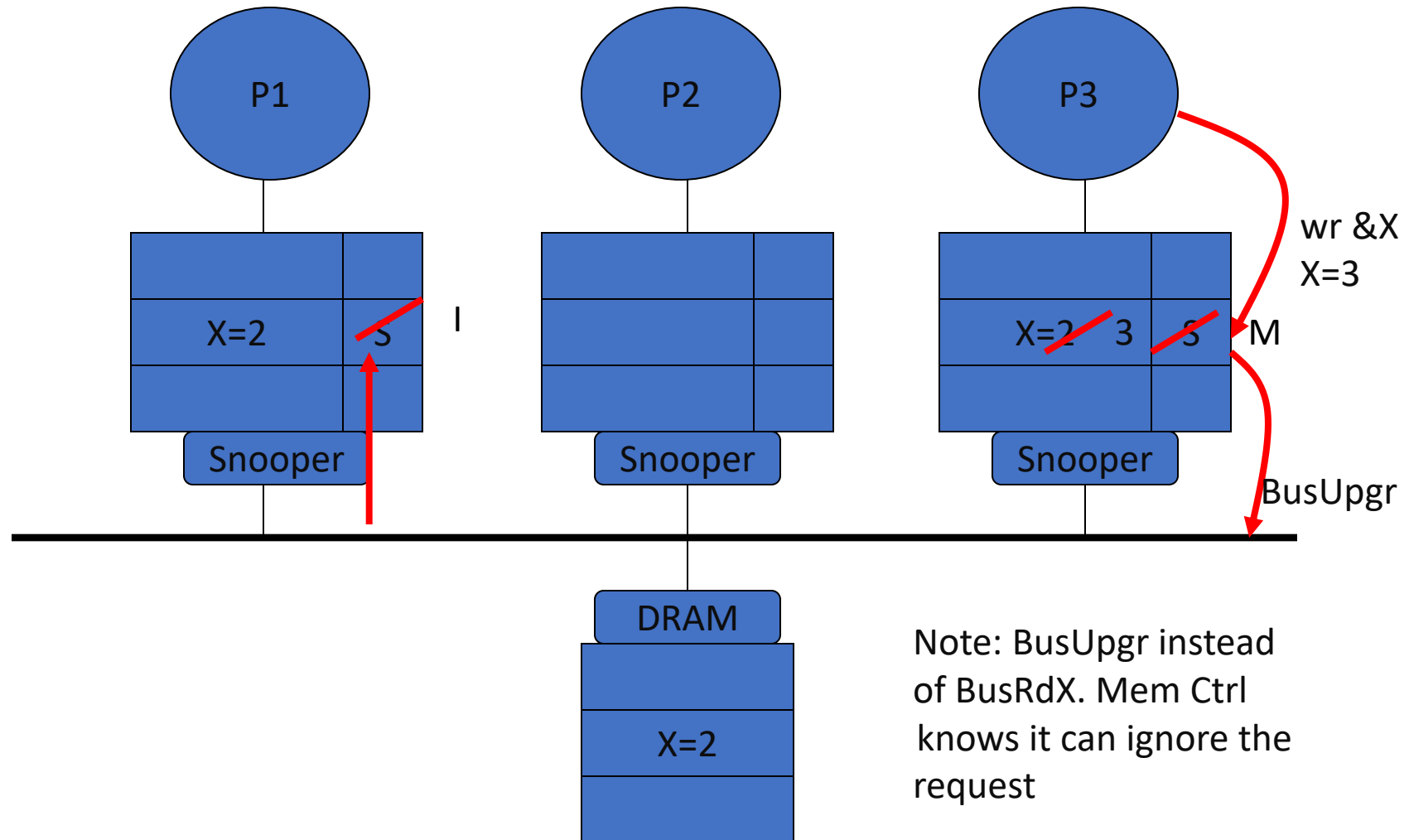
MESI



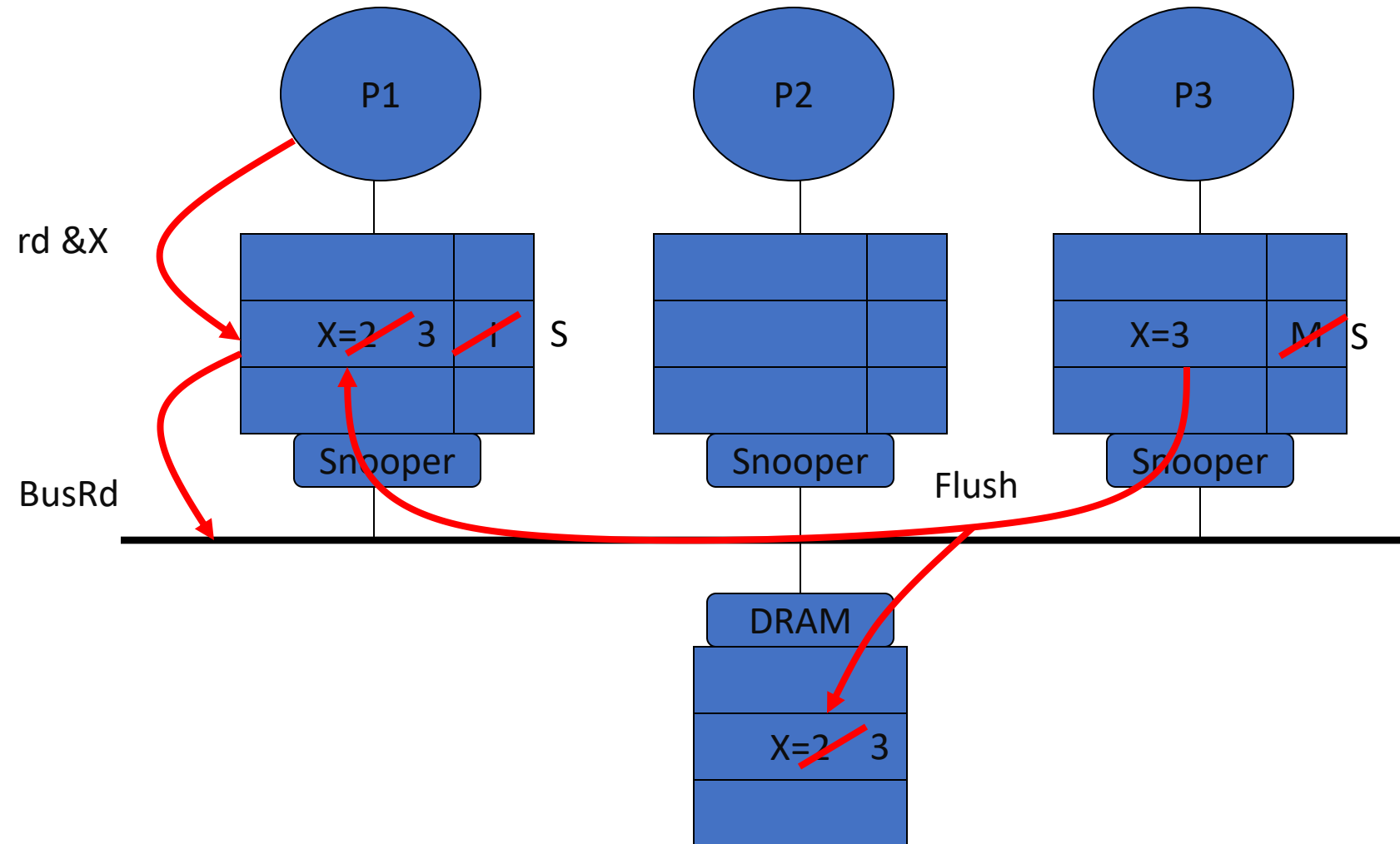
MESI



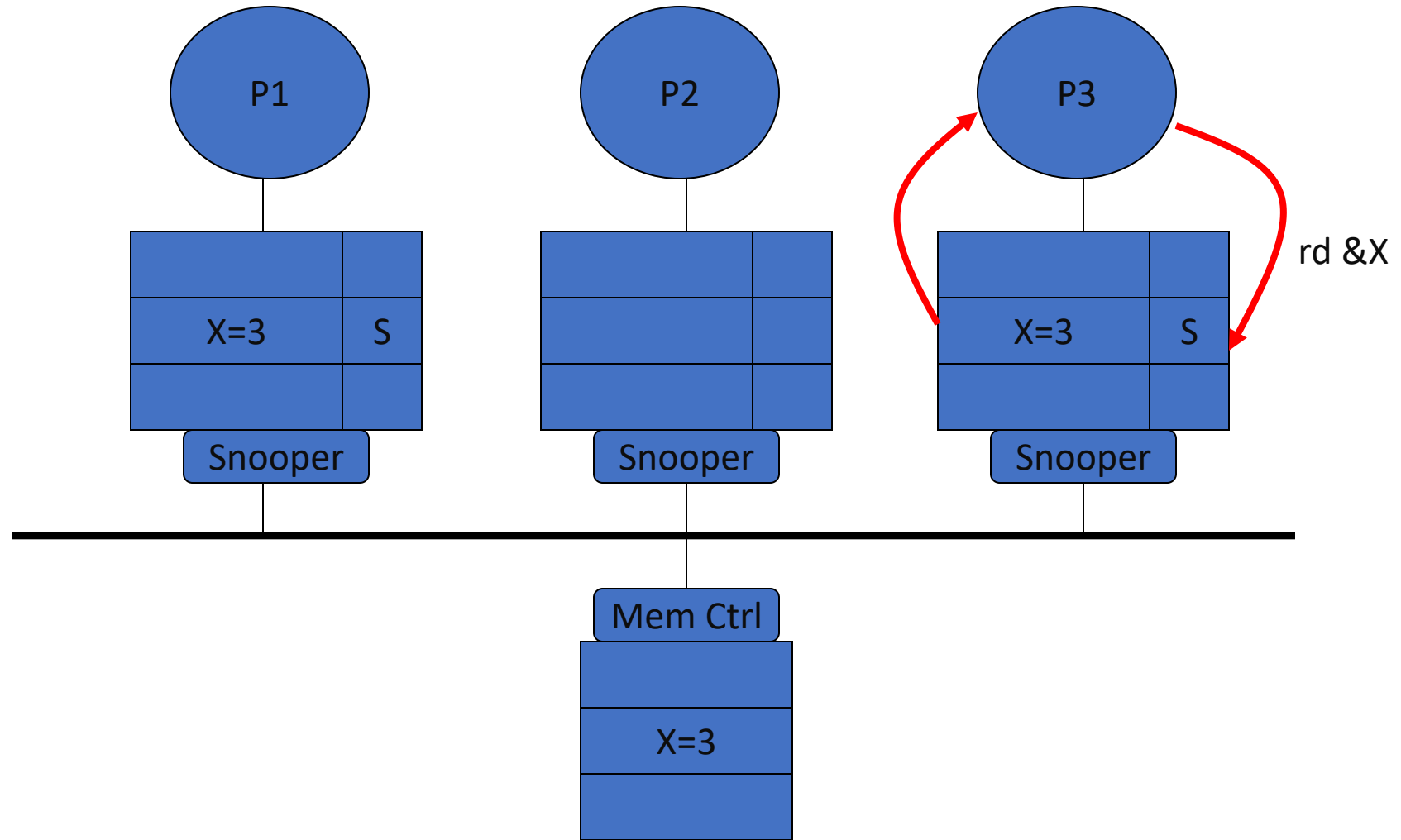
MESI



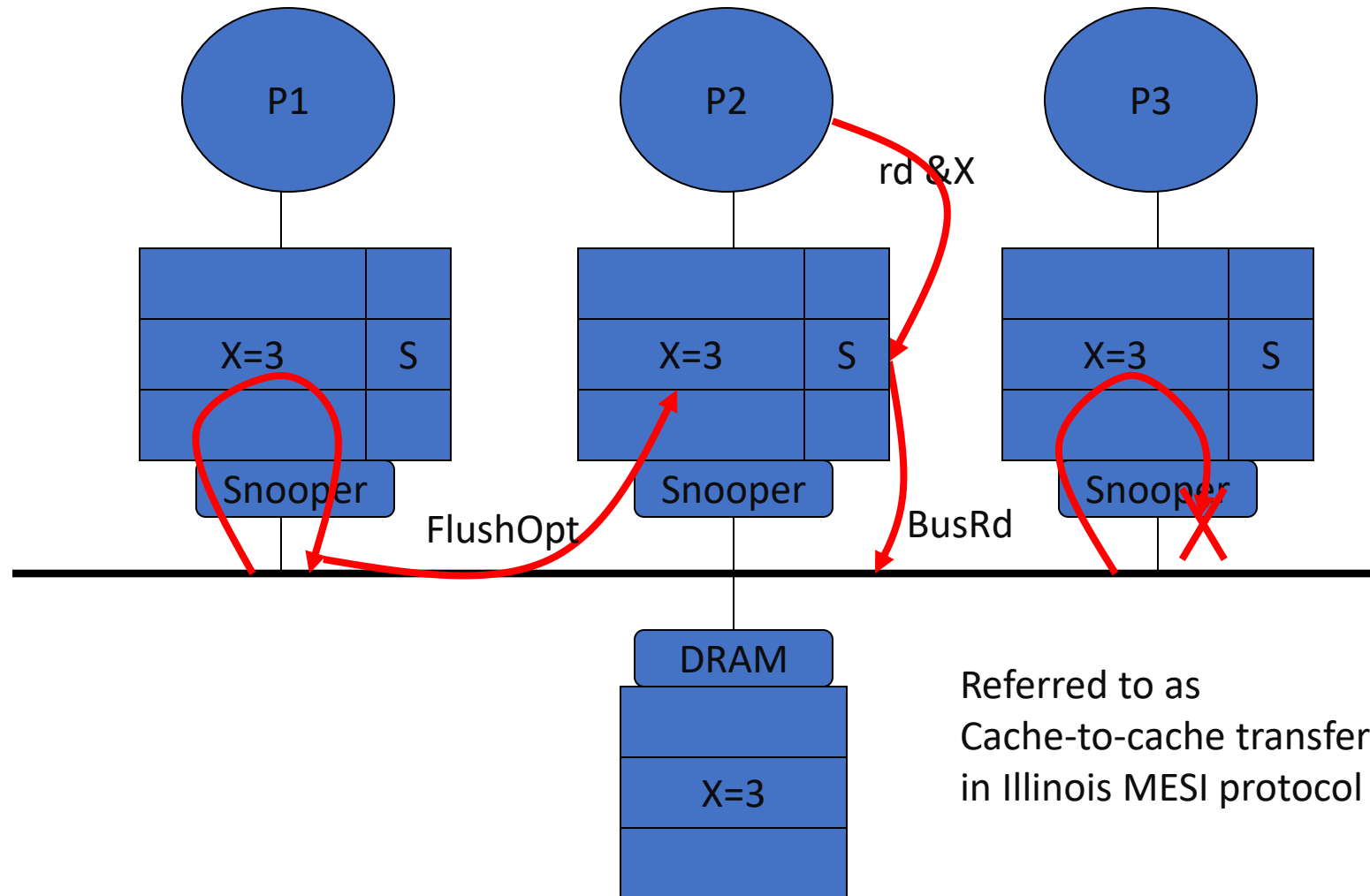
MESI



MESI



MESI



MESI – An Example

Proc Action	State P1	State P2	State P3	Bus Action	Data From
R1	E	-	-	BusRd	Mem
W1	M	-	-	-	-
R3	S	-	S	BusRd/Flush	P1's cache
W3	I	-	M	BusUpgr	-
R1	S	-	S	BusRd/Flush	P3's cache
R3	S	-	S	-	-
R2	S	S	S	BusRd/Flush Opt	P1/P3's cache

Cache Coherence Performance

Coherence misses: misses due to accesses to blocks that were invalidated due to coherence events

True sharing: misses that occur when multiple threads share the same data item (byte/word)

False sharing: misses that occur when multiple threads share different data items located in a single cache block

Cache Coherence Performance

Parameters	True Sharing Misses	False Sharing Misses
Larger cache size	increased	increased
Larger block/line size	decreased	increased
Larger associativity	unclear	unclear

Miss Latency for With coherence states for core-0

E State

Source	L1	L2	L3	DRAM
Local	4	10	38	65
Core 1 - 3	65	65	65	65
Core 4 - 7	186	186	186	106

M State

Source	L1	L2	L3	DRAM
Local	4	10	38	65
Core 1 - 3	-	-	38	65
Core 4 - 7	109	109	109	109

Miss Latency for Core 0

S State

Source	L1	L2	L3	DRAM
Local	4	10	38	65
Core 1 - 3	38	38	38	65
Core 4 - 7	170	170	170	106